bodor
Dare to dream

Laser Cutting Machine



C Series Product Analysis &
Instructions for Quick Learning

Overview

To begin with, it is a great honor to cooperate with you as a partner. What's next is a magical journey that laser processing technology brings to us. Let's enjoy it!

We believe this is just a start, and we genuinely wish that we can "Change Human Life with Laser Technology" together.

If this is your frst time knowing the mysterious power of laser, make sure to read the instructions before installation.

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Safety Notes

This Chapter explains the concept of safety, and indicates the measures that should be taken to avoid possible risks. The "Overview of Other Risks" includes the measures taken by users to avoid other risks.

There may be hazards and environmental factors when using this product. Among them, the hazard factors include power failure, non-standard use of gas cylinders and product cooperation, and environmental factors include noise pollution and waste disposal. In order to reduce energy consumption, the product also adopts energy-saving measures such as efficient design, energy saving mode and energy management to improve the realization of sustainable development.

Tip: Users must comply with the existing regulations of safety and accident prevention of each nation, as well as the safety regulations of each province and region.

Warning signs and danger signs:

Some actions may cause dangers during operation. Job description usually follows warning signs in the Instructions. Danger signs are set on the machine tool.

Warning signs include the following warnings:

Danger: Indicates there is severe harm. If it is not avoided, it may cause death or serious injury!!!

No.	Name	Label	Description
1	Anti-electric Shock Label		Warning: Indicates there is likely to cause dangerous situations. If it is not avoided, it may cause serious personal injury or great loss!



2	Careful extrusion label		Do not reach in during operation, and cut off the power before maintenance!
3	Prohibit users from inserting or removing optical fiber labels		Do not plug or unplug optical fibers!
4	Prohibit forklifts to insert labels		Prohibit forklifts to insert!
5	Shipping Mark		
6	Keep Upright Label		Please pay attention to this label on arrival and confirm that if the equipment is placed upside down.



7	Anti-reverse Label		Please pay attention to this label on arrival and confirm that if the equipment is seriously reversed or dumped.
8	Pay attention to the forklift label		The personnel are warned to pay attention to the forklift here.
9	Hoisting Label		The personnel are warned to pay attention to the hoisting location of the crane.
10	Emergency Stop Label		The personnel are warned to pay attention to the emergency stop button here.
11	Label of Removing the Shipping Block and Rust Prevention Oil		The personnel are warned that it is required to remove the rust prevention oil, shipping block and hoisting ring before using.
12	No Unpacking Label		The personnel are warned of not packing the boxes.



I. Overview

Please understand the content of this chapter before reading the manual.

This chapter shows you the intended readers, symbols, terms, abbreviations, expression rules and product conformity in the manual.

1.1 Scope of application

The manual is applicable to model C series fiber laser cutting machine.

1.2 Signal words

In the manual, the following safety signs identify important warning messages. When you see these sign, be alert to possible injuries, carefully read message on the right of the sign and inform other users in time.

The signal words are shown in Table 1-1 from high to low by risk level.



Table 1-1 Explanation of signal words

Sign	Signal word and explanation
DANGER	This signal word indicates a hazard with a high level of risk which, if not avoided, will result in death or serious injury.
WARNING	This signal word indicates a hazard with a medium level of risk which, if not avoided, could result in death or serious injury.
CAUTION	This signal word indicates a hazard with a low level of risk which, if not avoided, could result in minor or moderate injury.

When more than one risk level occurs, use the highest level. When a signal word contains a warning message that may cause personal injury, a warning message of possible property damage may also be attached.

1.3 Intended readers

The manual addresses the following skilled users:

- Delivery personnel
- Installation personnel
- Operator
- Maintenance personnel

WARNING	Read *Safety* instructions before working for all people allowed to use the machine



1.4 Terms

The terms and explanation used in the manual are shown in Table 1-2.



Table 1-2 Terms and explanation

No.	Term	Explanation
1	Plate machine	Machine for cutting rectangular plates, special-shaped plates and other plates.
2	Tube machine	Machine for cutting tubes such as round tubes and square tubes.

1.5 Abbreviations

The abbreviations used in the manual are shown in Table 1-3.

Table 1-3 Abbreviations

No.	Abbreviation	Full name
1	AC	Alternating Current
2	DC	Direct Current
3	HMI	Human Machine Interface
4	PLC	Programmable Logic Controller

1.6 Unit of measurement

The quantities in the manual are indicated in international standard units (SI Units). The units of measurement in the manual are the same as those on the machine.

1.7 Explanation of illustrations



For a better understanding of the manual, the illustrations in the manual are only for a brief description, in which the protective devices and covers may be removed. In the manual, Bodor Laser does not provide specific dimensions of the machine.

1.8 Conformity of machine

The machine is designed and manufactured according to the standards in Table 1-4.

Table 1-4 Reference standards

No.	Standard
1	Directive 2006/42/EC (Only CE standard equipment is applicable)
2	EN ISO 12100: 2010 Safety of machinery - General principles for design - Risk assessment and risk reduction
3	EN 60204-1:2018 Safety of machinery - Electrical equipment of machines Part 1: General requirements
4	Directive2014/30/EU (Only CE standard equipment is applicable)
5	EN IEC 61000-6-4:2019 Generic standards - Emission standard for industrial environments
6	EN IEC 61000-6-4:2019 Generic standards - Immunity for industrial environments
7	IEC 60825-1:2014 Safety of laser products - Part 1: Equipment classification and requirements
8	ISO 11553-1:2020 Safety of machinery - Laser processing machines - Part 1:



	Laser safety requirements



II. Safe operation of equipment

2.1 Basic safety rules

Most accidents happen because basic safety rules are not followed.

The safeguards provided by Bodor Laser are only for basic accident prevention, only when safety working procedures made by the proprietor are provided simultaneously, the machine can be operated and maintained safely.

The proprietor of the machine must ensure that:

1) The safety regulations are posted in the respective working area;
2) Those persons who are concerned have been trained for their particular job and in regard to the safety regulations;
3) The safety regulations are adhered to.

In order to give full play to the performance of the machine without personnel injury and machine damage, users must obey the following basic safety rules:

1) The machine must only be used for the purpose it has been designed for, refer to *2.3 Intended use and reasonably foreseeable misuse*. Do not use the machine exceeding the limits established by the technical data. Do not use the machine in a potentially flammable and/or explosive environment.



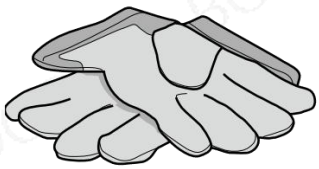
2) Operators must be qualified and authorized personnel who got safety and skill training. Maintenance of the machine should be carried out only by skilled persons who got specialized maintenance training of the machine. The work of the electrical system should be carried out by skilled electricians in accordance with applicable electrical engineering rules. These users should be able to fully comply with safety rules and preventive measures designed to prevent personal injury or machine damage.

3) Before using the machine, users must read the manual carefully, especially *Safety instructions* and all safety labels on the machine (refer to *Residual risks and safety labels*). Users must ensure that they fully understand the meanings and strictly follow its requirements, all recommended safety protection measures and "common sense", otherwise it will cause personal injury or machine damage.

4) For safety, wear appropriate personal protective equipments (PPE) in accordance with local safety regulations and company safety rules. When using the machine, the operator should wear safety shoes, protective gloves, protective glasses, protective masks, do not wear loose clothing, fasten the cuffs and tighten the hair. Do not wear rings, watches, bracelets, ties, scarfs and other accessories.









5) Before any operation, focus on what you want to do. Do not operate and maintain the machine after drinking or taking medicine. These conditions are critical to your safety.

6) Maintain good lighting conditions, a clean environment and sufficient working space to facilitate the operation and maintenance of machines and control cabinets.

7) Make sure that all safety protection measures are in place and maintained in effective working states before the operation, otherwise do not operate and maintain the machine, and immediately notify the superior. Do not suspend, move, modify or remove the safeguards or control devices of the machine at any time.

8) According to the noise test requirements specified in the EN ISO 11553-1-2020 standard, the A-weighted sound pressure of the main work stations of this machine is less than 70dB (A).

9) Before operation, the operator must place the board within the processing range of the equipment and below the bottom of the crossbeam to ensure that the crossbeam does not collide with the board when moving; When processing sheet metal, it is necessary to use a ruler as a reference to ensure that the sheet metal is always within the range of the table blade, in order to avoid safety accidents such as personal injury and equipment damage;

10) Personnel are not allowed to work on the workbench in principle. If there are special requirements, personnel need to wear protective gear and lay stepping steel plates on the workbench to avoid stepping on it. Before personnel enter the workbench, it is necessary to ensure that the laser cutting machine has been completely shut down and the power supply has been cut off. At the same time, the laser beam emitting device should be turned off to ensure no laser leakage.



2.2 Machine nameplate

The machine nameplate is shown in Figure 2-1.

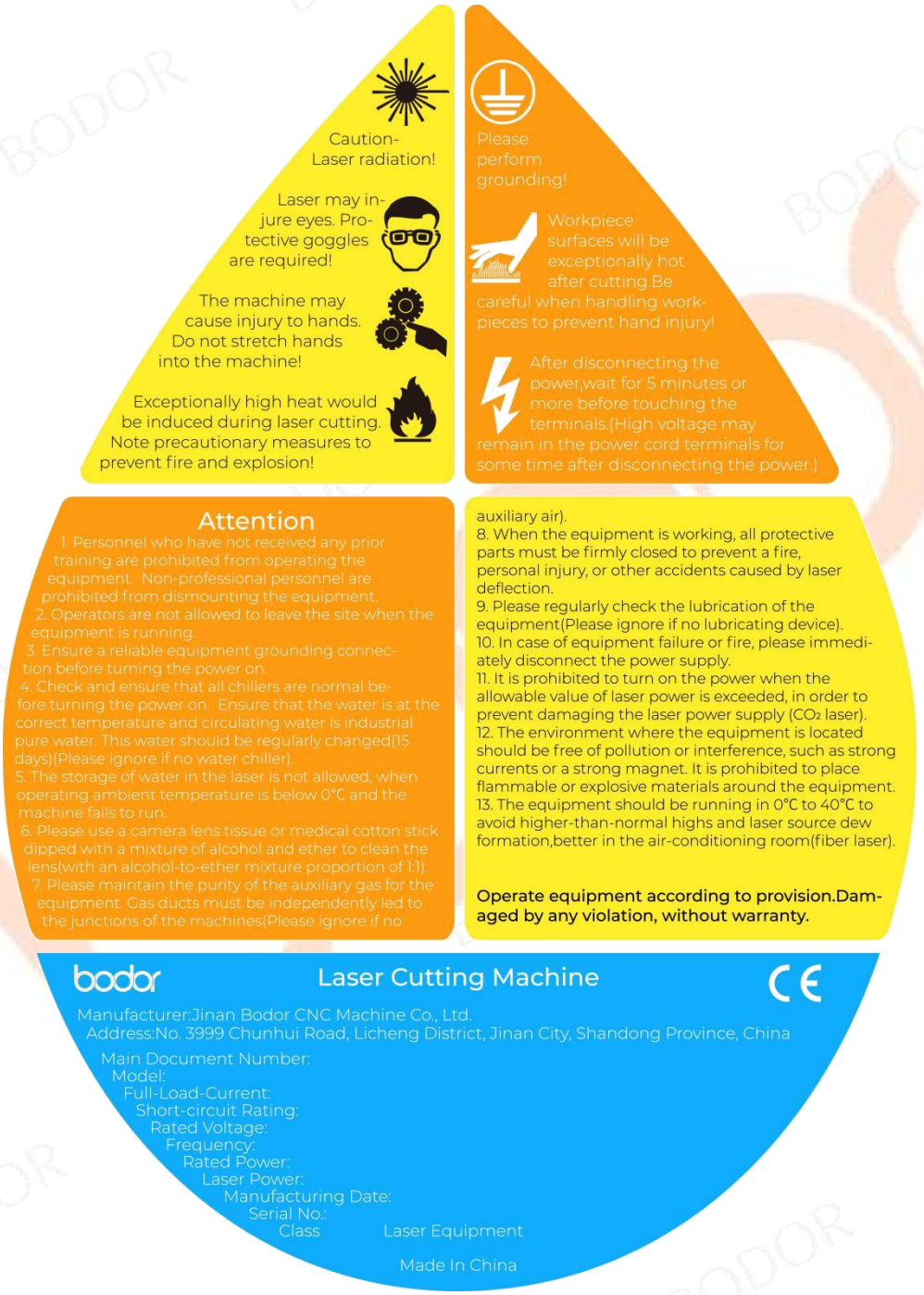


Figure 2-1 Machine nameplate



2.3 Intended use and reasonably foreseeable misuse

Laser cutting machines are designed and manufactured specifically for the cutting and processing of carbon steel, stainless steel, aluminum, and copper plates, as well as carbon steel and stainless steel pipes.

Machine users should fully understand the risks of the machine and have a strong sense of safety. Laser cutting machines are manufactured with the latest technology and safety standards. However, their use may cause harm to the lives and limbs of users or third parties, or damage to the machine and other material properties. Therefore, any faults that affect the safety of the machine should be immediately corrected.

Reasonably foreseeable misuse and solutions

Reasonably foreseeable misuse	Damage caused	Solution
The customer uses a non scanning cutting laser head to cut non-ferrous metals.	Non ferrous metals can generate high reflux during laser cutting, causing damage to the internal components of the laser head.	The user manual clearly stipulates that it is strictly prohibited to use non scanning laser heads to cut non-ferrous metal materials.
The user's cutting of the steel plate has exceeded the upper limit of the equipment's load-bearing capacity.	The AC motor is damaged due to excessive load, and the workbench is deformed due to excessive load.	The user manual clearly stipulates that different power segmented devices can cut different thicknesses of sheets.



The customer uses a laser cutting machine to cut materials such as glass, wood, rubber, etc.	Excessive laser temperature can cause burns to cutting materials, damage to equipment, and even threaten personal safety.	The user manual clearly stipulates that it is strictly prohibited to cut fragile and flammable materials such as glass, wood, rubber, etc

⚠ WARNING	Users must follow the requirements in the manual. It is dangerous to use the machine for function beyond intended use. In this case, users are fully responsible for the risk, Bodor Laser will not be responsible for any injury caused thereby. This includes but is not limited to the following: Do not use the machine exceeding the limits established by the technical data; Do not use machine beyond permitted materials, such as rubber, plastic, glass, wood, magnesium, and sheets with PVC film; It is prohibited to use dust collectors when cutting aluminum plates, magnesium aluminum plates, or other special plates, as the combustible dust generated by them poses a risk of fire or explosion. If customers have a specific need, explosion-proof dust collectors must be specially selected and equipped.

2.4 Operation positions

The operation positions of the machine are shown in Figure 2-2.

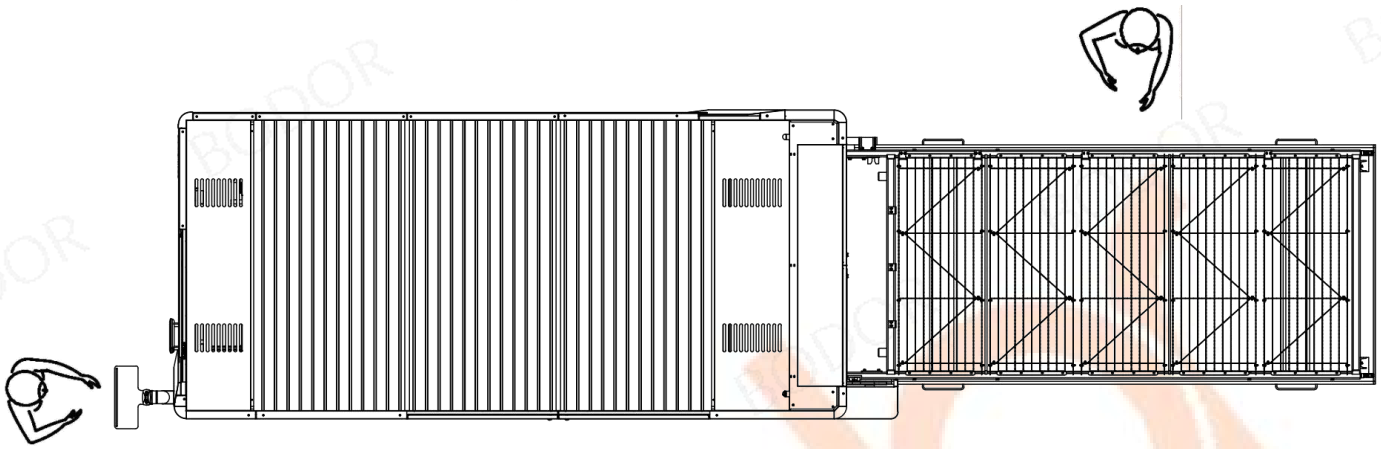


Figure 2-2 Operation positions

2.5 Operation environment

⚠ DANGER	The machine is limited to be used in the following environments! Operations beyond the environments may bring greater risks or affect the machine's function and performance.

1) Indoor use. Note: Avoid thermal deformation caused by direct sunlight on one side or wind on one side (for example, blinds can be installed when installing in the window position to avoid these situations).

2) Ambient temperature: 5℃～40℃。（For very hot and cold environments, additional requirements are required）.

3) Relative air humidity: ≤70%.

4) Atmospheric pressure: 86KPa~106KPa.



5) Vibration isolation, no corrosive medium, no strong magnetic field interference, dust degree ≤ 2mg/m3.

6) It is recommended that capable users provide a constant temperature and humidity environment for the equipment.

7) No conductive dust and corrosive gases that damage insulation.No vibration, shock and radiation affect the machine. When there are electric welding and electric discharge machines nearby, the equipment must be powered off and cannot continue to work, otherwise Bodor Laser will not be liable.

8) No potential flammable and/or explosive environment.

9) Make sure that there are no particles or substances that can absorb rays with a wavelength of 1.064μm around the machine. For example, solvent-containing steam emitted during painting, or steam emitted from degreasing equipment.

10) Avoid the water cooler environment close to the dew point to avoid equipment damage.

The dew point temperature refers to the temperature at which the air is cooled to saturation without changing the water vapor content and air pressure. To put it vividly, the temperature at which the water vapor in the air turns into dew is called the dew point temperature, as shown in Table 2-1.



Table 2-1 Comparison Table

Relative Humidity Ψ(%)	30	35	40	45	50	55	60	65	70	75	80	85	90	95
Environment Temperature Ta(℃)	Dew Point Td(℃)													
10	-7	-5	-3	-1.3	0	1.5	2.5	3.6	4.8	5.8	6.7	7.6	8.4	9.2
11	-6.5	-4	-2	-0.5	1	2.5	3.5	4.8	5.8	6.7	7.7	8.6	9.4	10.2
12	-5	-3	-1	0.5	2	3.3	4.4	5.5	6.7	7.7	8.7	9.5	10.9	11.2
13	-4.5	-2	-0.2	1.4	2.8	4.1	5.3	6.6	7.7	8.7	9.6	10.5	11.4	12.2
14	-3.2	-1	0.7	2.2	3.5	5.1	6.4	7.5	8.6	9.6	10.6	11.5	12.4	13.2
15	-2.3	-0.3	1.5	3.1	4.6	6	7.3	8.4	9.6	10.6	11.6	12.5	13.4	14.2
16	-1.	0.5	2.4	4	5.6	7	8.3	9.5	10.	11.	12.	13.	14.	15.



	3								6	6	6	4	3	2
17	-0.5	1.5	3.2	5	6.5	8	9.2	10.2	11.5	12.5	13.5	14.5	15.3	16.2
18	0.2	2.3	4	5.8	7.4	9	10.2	11.3	12.5	13.5	14.5	15.4	16.4	17.2
19	1	3.2	5	6.8	8.4	9.8	11	12.2	13.4	14.5	15.4	16.5	17.3	18.2
20	2	4	6	7.8	9.4	10.7	12	13.2	14.4	15.4	16.5	17.4	18.3	19.2
21	2.8	5	7	8.6	10.2	11.7	12.9	14.2	15.3	16.4	17.4	18.4	19.3	20.2
22	3.5	5.8	7.8	9.5	11	12.5	13.8	15.2	16.3	17.3	18.4	19.4	20.3	21.2
23	4.4	6.8	8.7	10.4	12	13.5	14.8	16.2	17.3	18.4	19.4	20.4	21.3	22.2
24	5.3	7.7	9.7	11.4	13	14.5	15.8	17	18.2	19.3	20.4	21.4	22.3	23.1
25	6.2	8.6	10.5	12.3	14	15.4	16.8	18	19.1	20.3	21.3	22.3	23.2	23.9
26	7	9.4	11.4	13.2	14.8	16.3	17.7	19	20.1	21.2	22.3	23.3	24.2	25.1
27	8	10.	12.	14	15.	17.	18.	19.	21.	22.	23.	24.	25.	26.



		3	2		8	3	7	9	1	2	2	3	2	1
28	8.8	11.2	13.2	15	16.7	18.1	19.6	20.9	22	23.1	24.2	25.2	26.2	27.1
29	9.7	12	14	15.9	17.6	19.2	20.5	21.3	23	24.1	25.2	26.2	27.2	28.1
30	10.5	12.9	15	16.8	18.5	20	21.4	22.8	23.9	25.1	26.2	27.2	28.2	29.1
31	11.4	13.7	15.9	17.8	19.4	20.9	22.4	23.7	24.8	26	26.9	28.2	29.2	30.1
32	12.2	14.7	16.8	18.6	20.3	21.9	23.3	24.6	25.8	27	28.1	29.2	30.1	31.1
33	13	15.6	17.6	19.6	21.3	22.9	24.2	25.6	26.8	28	29	30.1	31.1	32.1
34	13.9	16.5	18.6	20.5	22.2	23.8	25.2	26.5	27.7	29	29.5	31.1	32.1	33.1
35	14.9	17.4	19.5	21.4	23.1	24.6	26.2	27.5	28.7	29.9	31	32.1	33.1	34.1
36	15.7	18.1	20.3	22.2	24	25.7	27	28.4	29.7	30.9	32	33.1	34.1	35.2
37	16.6	19.2	21.2	23.2	24.9	26.5	27.9	29.5	30.7	31.8	33	34.1	35.2	36.2
38	17.	19.	22	23.	25.	27.	28.	30.	31.	32.	33.	35.	36	37



	5	9		9	8	4	9	3	5	7	9	1		
39	18.1	20.8	23	24.9	26.6	28.3	29.8	31.2	32.5	33.8	34.9	36.2	36.8	38.1
40	19.2	21.6	23.8	25.8	27.6	29.2	30.7	32.1	33.5	34.7	35.8	36.8	38.1	39.1

Bodor recommends the following measures to extend the service life of the laser:

1) In order to ensure a good operating environment of the laser, reduce the risk of condensation, reduce the frequency of failure and extend the service life. For the equipment with power of 10000 watts and above, the customers are required to configure the air-conditioned room of the laser. The air-conditioned room should be spacious enough for easy access to internal maintenance. The minimum clearance around the laser is 1.0m. The air-conditioned room is required to have good thermal insulation and sealing performance to provide a constant temperature and humidity environment for the laser, and the temperature in the air-conditioned room is required to be ≤ 28 ℃. Relative humidity ≤ 50%.
2) The water cooler and the laser should be separately placed in the unused space. It is strictly forbidden to place the water cooler in the air-conditioned room of the laser.

Disclaimer: the laser is damaged due to the customer's failure to configure the air-conditioned room for the laser. The loss and various expenses caused by this are borne by the customer and are not within the scope of bond laser's responsibility.



2.6 Energy requirements

2.6.1 Electrical power

The electrical power requirements of the machine are shown in Table2-2.

Table 2-2 Electrical power requirements

Voltage	Nominal voltage AC380V Steady state voltage: 0.9 to 1.1 of nominal voltage.
Grounding system	3P+N+PE
Frequency	Nominal frequency 50Hz, 0.99 to 1.01 of nominal frequency continuously, 0.98 to 1.02 short time.
Harmonic	Harmonic distortion not exceeding 12% of the total r.m.s. voltage between live conductors for the sum of the 2nd through to the 30th harmonic.
Unbalanced voltage	Neither the voltage of the negative sequence component nor the voltage of the zero sequence component in three-phase supplies exceeding 2% of the positive sequence component.
Voltage interruption	Supply interrupted or at zero voltage for not more than 3 ms at any random time in the supply cycle with more than 1000ms between successive interruptions.
Voltage drop	Voltage dips not exceeding 20 % of the rms voltage of the supply for more than



	one cycle with more than 1000ms between successive dips.
Surge protective devices (SPDs)	SPDs should be provided in the power supply system by end-user, to protect against the effects of overvoltages due to lightning or switching surges.

If the power quality does not meet the above configuration, a voltage stabilizer must be added to ensure normal operation of the equipment. Bodor Laser is not liable for losses caused by the quality of the power supply, especially property losses, personal injuries and production losses. Such risks are entirely borne by the user.

The configured transformer/stabilizer is only used for this machine, and it is forbidden to supply external power to other equipment.

2.6.2 Air supply

⚠ CAUTION	Laser cutting machine needs to be equipped with a separate air compressor.

The air pressure machine should be installed in a clean, dry, well -ventilated room, without dust and harmful gases. The temperature of the working environment is not higher than 45℃. The relative humidity of the surrounding atmosphere should be less than 80%. The installation of the ground must be solid and flat. And maintain the level. If the working temperature is lower than 0℃, you need to pay special attention to whether the air compressor drainage is frozen. For details, please carefully read the external identification sticker of the air compressor 《Precautions for Use》or the air compressor instructions.

The requirements of the air compressor are shown in Table 2-3.



Table2-3 Air compressor requirements

	Low power	High power	Ultra high power
Pressure	1.6MPa	1.6MPa	1.6MPa
Flow	1.5m^3/min	2.6m^3/min	3.7m^3/min
Pressure dew point	2 ~ 8℃	2 ~ 8℃	2 ~ 8℃
Filtration accuracy	0.01μm	0.01μm	0.01μm
Oil content of air outlet	≤0.003ppm	≤0.003ppm	≤0.003ppm

The gas quality complies with ISO 8573-1:2010 standard requirements, solid particles-grade 2, water-grade 4, and oil-grade 3. It is required to be clean, free of water, oil and other impurities. It is used for thin carbon steel and thin stainless steel where conditions permit.

Disclaimer: The use of compressed air below the above requirements will cause irreversible pollution to the machine tool pipeline, air valve and cutting head. The resulting losses and various costs are borne by user and are not within the scope of Bodor Laser's responsibility.



2.7 Residual risks and safety labels

2.7.1 Residual risks

The residual risks of the machine are as follows.

⚠ WARNING	1) The rupture of pneumatic system air pipe will cause actuator such as cylinder to move. 2) Even if the main power switch is in the "OFF" position, the power supply line and main power switch incoming terminals are still live. 3) Even if the main power switch is in the "OFF" position, servo drivers, frequency converter and switching power supply may still be live 4) Burn hazards arising from the surface temperature of servo driver and frequency converter heatsink. 5) The surface temperature of the newly cut workpiece is high, which may cause burns.

2.7.2 Safety labels

Safety labels give informations of the machine's potential risks. Before operation, be familiar with all safety labels.

The layouts of safety labels are shown in Figure 2-3.

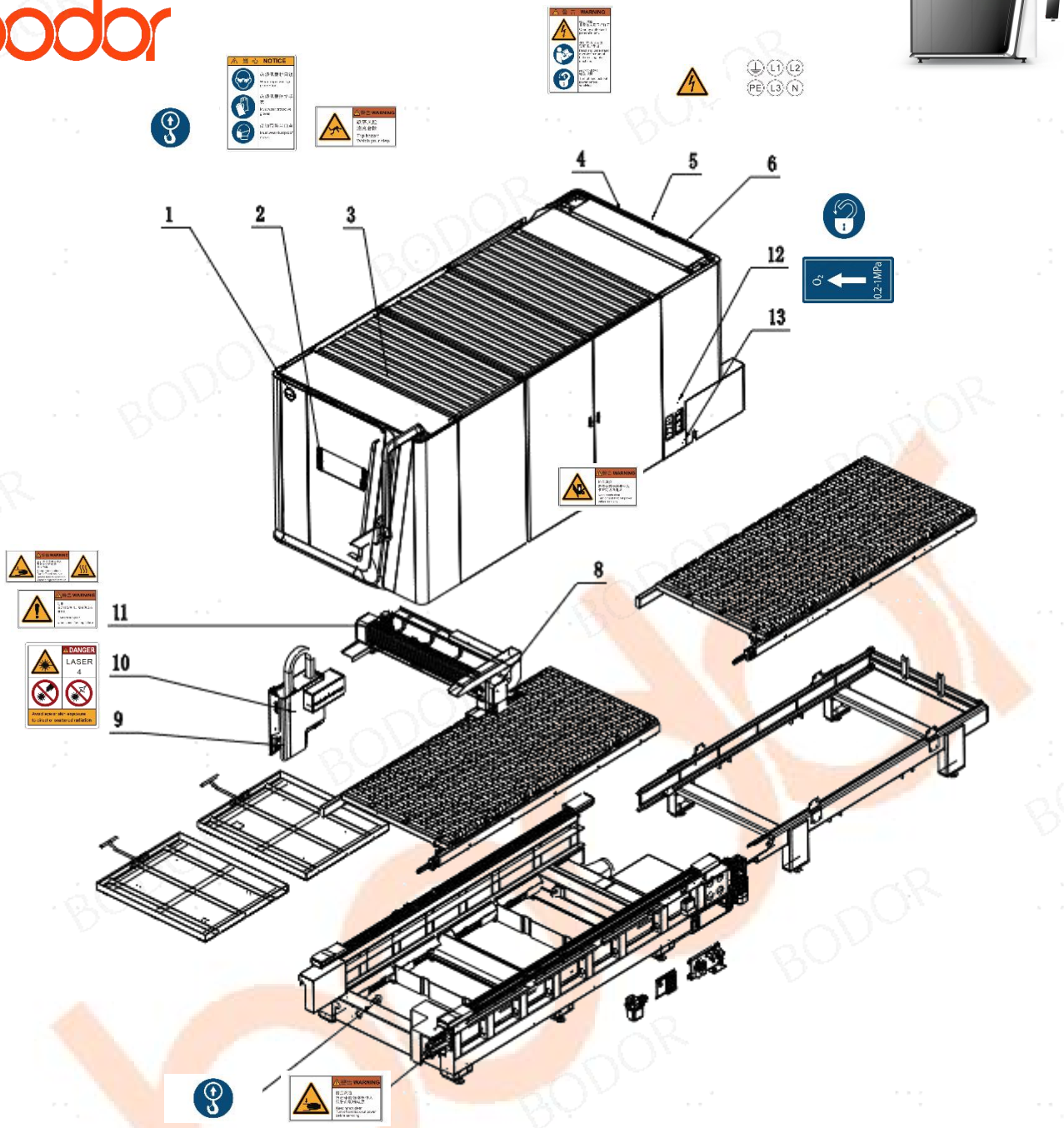


Figure 2-3 Safety labels layout

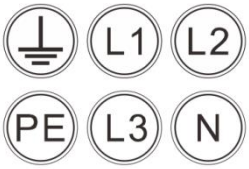
The descriptions of safety labels are shown in Table 2-4.



Table 2-4 Safety labels

No.	Position	Safety label	Description
1	Lifting point		Machine lifting point
2	Cover observation window		Class 1 laser products. Laser beams available are not dangerous under predictable proper conditions. Must wear approved eye protection. • Must wear approved eye protection. • Must wear protective gloves. • Must wear dustproof mask.
3	Trolley at bottom of cover		When the trolley is pulled out, there is a tripping hazard.
4	Control cabinet door		Open by authorized personnel only. Read and understand operator's manual before using this machine. Turn off and lock out power before servicing.
5	In control cabinet		Risk of electric shock



6	In control cabinet		Wiring instructions
7	Sprocket of exchange platform		Squeeze risk Do not put your hand in during operation Cut off the power before maintenance
8	Beam		Squeeze risk Do not put your hand in during operation Cut off the power before maintenance
9	Laser head		Class 4 laser products Avoid eye or skin exposure to direct or scattered radiation
10	Laser head		Contact supplier when Laser Tooling failure
11	Motor base		Burn hazard Hot surface Do not touch
12	Air source position		Turn off and lock out air power before servicing
13	Air source position		Air pressure and flow direction



Make sure that all safety labels are clear and legible. Clean or replace the safety labels if you cannot read.

When cleaning safety labels, use a cloth, soapy water. Do not use solvents, gasoline, or other harsh chemicals. Solvents, gasoline, or harsh chemicals could loosen the adhesives that secures the safety labels.

Replace any damaged or missing safety labels. If safety labels are attached to the part to be replaced, paste same safety labels on the new replacement part.

2.8 Safeguards

The safeguards are shown in Figure 2-4.

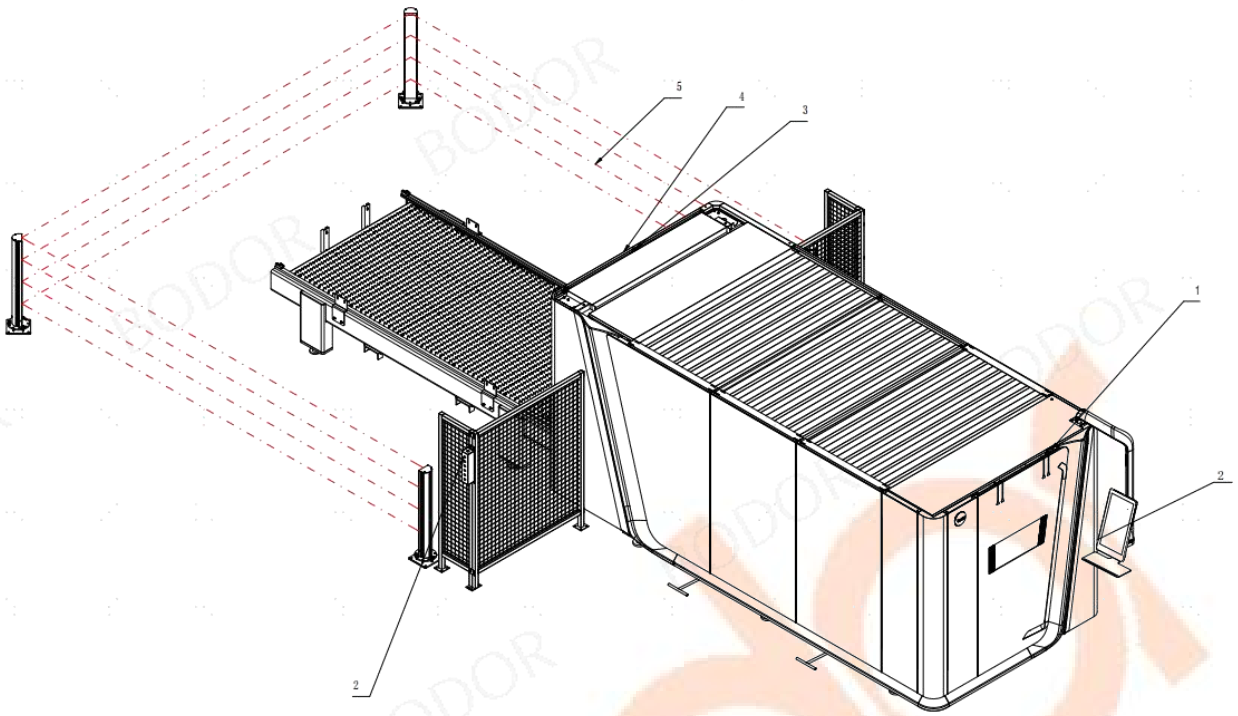


Figure2-4 Safeguards layout

The function descriptions of safeguards are shown in Table 2-5.

Table 2-5 Safeguards functions

No.	Safeguard	Description
1	Safety door switch	Open safety door, machine stops urgently. Close safety door and press "RESET" button on side of touch screen to reset safety system.
2	Emergency	Press emergency button and machine stops urgently. Rotate to release



	button	emergency button and then press "RESET" button on side of touch screen to reset safety system.
3	Safety door switch	Open safety door, machine stops urgently and power off.
4	Main power switch (with keyhole)	Turn on/off main power of machine, turn off main power and lock it before maintenance.
5	Safety light curtain	The safety light curtain is triggered, and the table top exchange is stopped in an emergency. Press the "Reset" button on the side of the touch screen to restore the security system.

Please note that in safety protection devices, safety fence, light curtain and corresponding emergency stop button are only CE/CU standard configuration, GB is optional. During maintenance and repair of the electrical control cabinet, it is prohibited to step on the countertop. A dedicated maintenance platform (maintenance ladder) is required, refer to Appendix 1 of the instruction manual.



Safety instructions related to safeguards:

DANGER	1) Make sure that safety protection measures are in place and maintained in effective working state before operation. 2) Do not suspend, move, modify or remove the safeguards or control devices of the machine at any time.
WARNING	1) The safety protection measures taken by the machine can only reduce but not completely eliminate all risks. 2) Check the functions of emergency button and safety light curtain every 72 hours. 3) Check the functions of main power switch and safety door switch every 168 hours. 4) After any maintenance work of removing safety door or other safety protection devices, reinstall these devices, check whether they are functioning properly, and ensure that they are in the correct position and in an effective working state. 5) Check the necessary safety functions after replacing any safeguards. 6) Operator should know the position and operation method of emergency stop devices to stop the machine in case of emergency. 7) Do not stop the machine during usual operation by emergency stop devices.



2.9 Safety instructions for machine lifecycle

During the transportation, installation, operation, adjustment, maintenance and disassembly of the machine, users should obey the basic safety rules described in the manual, the safety regulations in your country/region and safety preventive measures as follow.

Disclaimer: Any operation outside of this regulation is regarded as a non-compliance use, and Bodor Laser is not responsible for the losses caused thereby, especially property losses, personal injuries and production losses. Such risks are entirely borne by the user.



2.9.1 Transportation

For personal safety and machine transportation safety, delivery personnel should obey the following safety rules besides 2.1 Basic safety rules:

1) Provide suitable storage conditions to prevent machine damage. The storage temperature should be between -25 ~ + 55℃, and for short periods not exceeding 24 h at up to +70℃.
2) Only qualified and authorized personnel who got safety and skill training can transport the machine;
3) Refer to machine mass shown in 5.3 Transportation and use lifting equipment with sufficient capacity. Only use the lifting points shown in 5.3 Transportation.
4) Avoid sudden movement during machine moving.

2.9.2 Installation

For personal safety and machine transportation safety, installation personnel should obey the following safety rules besides 2.1 Basic safety rules:

1) Only qualified and authorized personnel who got safety and skill training can install the machine;
2) Before installation, check if

- No part of the machine is damaged;
- No water leakage in the control cabinets, operation panels and junction boxes;
- Internal pipeline connection is firm and mechanical connection is reliable.



3) The machine should be reliably grounded. Before connecting the live conductors of power cable to the machine, ground the machine first. The plate machine is grounded separately, tube machine is grounded separately, control cabinet is grounded separately, and peripheral equipment (laser, water cooler, transformer, voltage stabilizer, fan, dust collector, etc.) is separately grounded. Grounding resistance is required to be ≤4Ω;
4) PE terminal, the protective conductors in the equipment of the machine, conductive structure parts and exposed conductive parts and conductive structure parts of the machine should be interconnected.
5) Connect the protective bonding circuit to accessible extraneous conductive parts (such as metallic pipes, fences, ladders and handrails) in the vicinity of the machine.
6) After installation, verify the continuity of the protective bonding circuit by professional personnel equipped with test instruments.
7) Static elimination device shall be provided by the end-user for such material roll which undesirable electrostatic may discharge.

2.9.3 Operation and adjustment

For personal safety and machine operation safety, operators should obey the following safety rules besides 2.1 Basic safety rules:

1) Only qualified and authorized personnel who got safety and skill training can operate and adjust machine. This applies not only to the machine operator, but also to all those who may be involved in the machine operation in some way.
2) The key of the mode-select key switch should be kept by authorized person in charge.



3) The HMI password should be kept by authorized person in charge.

4) Any function adjustment of the machine must be performed when the machine is stopped.

5) Always keep hands, arms or other body parts away from moving parts of the machine. Do not reach into areas of dangerous moving parts or use tools to increase the accessibility of the areas.

6) Never climb or rest against the machine at any time.

7) Before starting the machine, make sure none is working on or around the machine and remove all irrelevant items on the workstation.

8) The machine should be operated by a single person.

9) Do not handle material over 25kg by a single person. When the mass of material is greater than 25kg, two persons lift together or use auxiliary equipment which should be provided by the user.

10) When changing the material from aluminum/aluminum alloy to other metal materials (and vice versa), you must wait for the dust removal device to run for a period of time before it can work. It is best to clean the filter element in the fume purifier once (if the equipment is equipped). When conditions permit, special filter elements are used for each type of material.

11) Use suitable tools to take out high temperature workpieces.

12) In case of serious failure of the water chiller (such as leakage of the coolant circulation circuit), the water chiller must be shut down immediately.

13) Do not use machine when dust removal system is not working.



14) Place a dry powder fire extinguisher next to machine to prevent fire hazards caused by thermal residues generated during laser processing; place a CO2 fire extinguisher to prevent fires caused by other reasons.

15) Do not allow cigarettes or other burning materials to enter the inside of the machine.

16) Do not allow cigarettes or other burning materials to enter the inside of the machine.

17) After emergency stop or failure, the machine should be checked by authorized personnel before restarting.

18) After stopping the machine under safe conditions, remove debris or other foreign matters from the machine and nearby.

19) Where whole body acess is possible, can enter, person who leaving danger zone needs to look that there is no person in the danger zone and to operate the reset switch before starting the machine.

20) In order to ensure the safety of human machines, in the manual mode, the X/Y axis operation speed is recommended to operate the equipment at the factory setting at the factory (low -speed 6m/min, high -speed 18m/min) operation equipment; if you need to adjust, please contact our post -sales service engineer and ensure the equipment The environment is safe within the operation and the surrounding environment.

21) After the production is finished, stop the power supply.



2.9.4 Maintenance

For personal safety and machine maintenance safety, maintenance personnel should obey the following safety rules besides *2.1 Basic safety rules*:

1) Only qualified and authorized personnel who got safety and skill training can do maintenance work.
2) Sufficient local lighting must be provided if maintenance or inspection requires access into the machine.
3) Maintenance signs must be placed around the maintenance area to warn the maintenance is in progress.
4) Before starting any maintenance work, make sure that the machine is completely stopped, turn main power switch to "OFF" position and lock it, and turn main air source switch to "OFF" position and lock it.
5) Do regular maintenance and inspections of machine in accordance with the requirements in 8 Maintenance.
6) Even if the main power switch is in the "OFF" position, the following parts and their conductors are still live!

- Power supply line and main power switch incoming terminals are still live.
- Servo drivers, frequency converter and switching power supply may still be live within 20 minutes after power off. Do not touch their parts and the conductors connected to them.

7) Do not move the positions of safeguards. Otherwise, Bodor Laser will not be responsible for repair or any damage caused thereby.



8) Take safety preventive measures in the manual to prevent personal injury or machine damage when doing maintenance work.

9) Reinstall the safeguards and make sure them in correct position and effective working states after removing the safeguards for maintenance purpose.

10) Use the parts and materials confirmed by Bodor Laser when replacing new parts.

11) Any technical modification of the machine must be done by Bodor Laser or with the confirmation of Bodor Laser.



2.10 Laser safety

Laser equipment is broken down into different laser classes according to the European standard EN 60825-1, laser class indicates the hazard level of laser beam irradiated, as shown in Table 2-6.

Table 2-6 Laser classes

Laser class	Description
1	Laser beams available are not dangerous under predictable proper conditions.
2	The wave length of laser beams available is within the visible spectrum range (400nm-700nm). Short-time exposure (less than 0.25s) does not damage the eyes. For other rays beyond the wave length range of 400-700nm which meet Class 1 conditions, generally can protect the eyes by keeping away from laser beams or closing eyes.
3B	Laser beams available are harmful to the eyes and skin as well. It is very dangerous to look straight at rays of a 3B laser. Exceeding the maximum allowable radiation value, rays from the 3B laser source may cause damage to the skin.
4	Rays available are extremely harmful to the eyes and skin, and even diffused rays also cause damage. And rays may cause a fire and explosion hazard. For a class 4 laser device, safety measures must be taken: The most important is to fully protect the eyes. Under normal circumstances, the power intensity of a Class 4 laser device is enough to burn the skin, cause a fire and ionize the atmosphere when focusing. Therefore, a series of other security measures should be taken.



The laser level is different under different machine operating modes. The operating modes of the laser cutting machine are shown in Table 2-7.



Table 2-7 Operating modes

Mode	Description	State	Class
Normal operation mode	Normal operation mode refers to the operation of the equipment within the range of all functions.	Machine is controlled by program or manually. Personnel are outside the danger zone.	1
Maintenance and debug mode	Maintenance and debug mode refers to state of machine during maintenance and debugging.	When the equipment is in an abnormal state or needs to be maintained.	4

The danger of laser rays.

DANGER	1) The laser beam in operation is high-energy, invisible light! 2) It may severely burn the skin and hurt the eyes. The impact on vision can cause vision damage and even blindness! 3) If you are debugging while the laser is working, you must wear laser protective glasses. 4) Thermal residues generated during laser processing can easily cause fire hazards.
WARNING	1) Laser rays can hurt eyes! Do not look directly at the rays! 2) If you look at the processing position for a long time, visible rays may



	cause injury to the operator. Especially in high-speed cutting, or processing such as galvanized steel, aluminum and high-quality steel, the plasma generated will produce bright and strong light, which may cause permanent damage to the retina. 3) The secondary rays during laser processing are dangerous! High-intensity visible rays can cause permanent damage to the retina! 4) Do not look directly at the processing location without taking protective measures 5) Wear laser protective glasses when needed. 6) Cutting highly reflective materials can easily cause fire and may damage the laser and fiber, it is not recommended to cut for a long time.



Disclaimer:

1) Any operation other than those regulated herein shall be considered as non-compliant use; any losses caused thereby, especially property damage, personal injury and production losses, shall be fully borne by the user.
2) Subject to the risk of laser-bounce during the laser cutting process, there should be no flammable and explosive products such as gasoline and styrofoam within 2 meters around the laser cutting machine to prevent fires, explosions and other disasters. Any losses caused thereby, especially property damage, personal injury and production losses, shall be fully borne by the user.
3) If the weight of the plate per unit area placed in any position of the equipment workbench exceeds the load per unit area of the workbench (unit area load = worktable load (kg) / processing width (㎡)), please contact Jinan Bodor Laser for assistance in providing solutions. Any operation beyond this regulation will be regarded as illegal operation, and Bond Laser will not be liable for any damage to the equipment, personal and property losses caused thereby.
4) The protective devices and functions of the equipment itself must not be removed or discarded by the customer without permission. Any losses caused by the failure to observe this requirement, especially property damage, personal injury, production losses, shall be fully borne by the user.

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